



Article

Validation of a Web App Enabling Children with Dyslexia to Identify Personalized Visual and Auditory Parameters Facilitating Online Text Reading

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Abstract: Previous research has shown the importance of font type, size, and spacing to facilitate text reading in dyslexia. Great heterogeneity in the population of readers with specific learning disorders suggests that personalized parameters should be preferable compared to one-fits-all ones. A special automatized procedure was designed to select the most favorable parameters for both text visualization and text-to-speech conversion. A total of 78 primary and middle school students (29 typical readers, 49 children with atypical reading skills, either diagnosed as specific reading disorder or as special learning needs) took part in this study, which included the application of the procedure and a validation of its outcomes through a systematic comparison of the use of the personalized versus standard fonts and voices in reading and writing tests. The results show a significant advantage for the personalized parameters. Moreover, in the case of text-to-speech personalization, the advantage is significantly larger for dyslexic readers than for typical readers. These results confirm the usefulness of a personalization approach in providing support to facilitate learning in dyslexic students.



Citation: Lorusso, M.L.; Borasio, F.; Panetto, P.; Curioni, M.; Brotto, G.; Pons, G.; Carsetti, A.; Molteni, M. Validation of a Web App Enabling Children with Dyslexia to Identify Personalized Visual and Auditory Parameters Facilitating Online Text Reading. *Multimodal Technol. Interact.* **2024**, *8*, 5. <https://doi.org/10.3390/mti8010005>

Academic Editor: Andreas Riener

Received: 25 November 2023

Revised: 6 January 2024

Accepted: 9 January 2024

Published: 15 January 2024



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Keywords: dyslexia; personalization; font; spacing; size; web app; online reading; visualization; text-to-speech; pitch

1. Introduction

1.1. Background: Developmental Dyslexia and Perception

Developmental dyslexia (DD) is one of the most common neurodevelopmental disorders diagnosed in children who fail to develop normal reading skills in spite of normal intelligence [1]. A recent study conducted in Italy [2] showed a prevalence of 3.5%. Traditional approaches considered DD as a phonological processing impairment [3,4]; however, several other functions were shown to be specifically impaired, such as long-term and short-term verbal memory or working memory, visual attention, and visual and auditory perception [5–8].

The attentional systems that form part of the magnocellular system (i.e., the dorsal system or dorsal visual stream), control the ability to concentrate visual attention in a restricted area of the visual field and to shift it when needed. They also control so-called visual crowding, an automatic mechanism of our perceptual system producing a sort of “blurring” of the visual areas surrounding the target object, resulting in the masking of further visual elements present in those areas [9,10]. The visual attention systems play a role in the early stages of grapheme-to-phoneme conversion, and they were shown to be involved in visual searches and in graphemic parsing through neuronal-oscillation modulation mechanisms very similar to those involved in phonological processing [11,12]. The Temporal Sampling Framework (TSF) proposed by Goswami et al. [11] could be applied

to the different stages of processing within the visual system as well, before the stage of phonological processing. Visual crowding makes stimulus identification more difficult: in the case of text, single letters may be considered as target stimuli, and crowding effects can be observed also in central vision [13]. Visual crowding thus leads to the inability to recognize letters when surrounded by other letters, resulting in slower reading and mistakes [12,14,15].

More generally, Bouma's law of crowding describes an uncrowded central window through which we can read and a crowded periphery through which we cannot [16]. Reading rate is determined by crowding and eccentricity. During text reading, typical readers are limited by letter spacing much more than by font size [17]. Children with DD, moreover, show reduced accuracy for letter identification in and near the foveal field in comparison with typical readers, and a peculiar spatial distribution of lateral masking across central and peripheral vision [9,18].

1.2. Dyslexia-Friendly Fonts and Text-to-Speech Technologies

Over the past years, research has been looking for solutions to alleviate deficits in the visuospatial processing of letters and words [19], and to improve reading in individuals with DD using specific fonts. A study by Joo and colleagues [20], for instance, showed that individuals with DD read faster when a text is presented with increased inter-letter and inter-word spacing. An extra-large space (inter-letter spacing enlarged by 2.5 pt. on 14 pt. body size, +~18% of the body size) was found to improve reading in children with DD when they use the most common fonts, such as Times New Roman or Arial (regular space between two lowercase letters may vary roughly from 0 to 15% of the body size, while the regular space between two words corresponds approximately to 20–25% of the body size) [21,22]. Moreover, using sentences of the same length and extra-large inter-letter spacing was found to decrease the number of errors without increasing reading speed [21]. Also, Perea et al. [23] found that young readers with dyslexia showed faster reading times and higher comprehension scores when the text had a small increase in inter-letter spacing relative to the default settings. Non-dyslexic children and older children had reduced effects compared to younger children with dyslexia. A recent study by Łuniewska et al. [24] by contrast showed that, in both children with DD and typical readers, increased inter-letter spacing did not affect reading accuracy, speed, or comprehension but led to shorter fixations in dyslexic children with respect to normal readers, suggesting possibly different reading strategies.

Turning to font types, it is usually suggested that sans-serif fonts are easier to be read by people with DD than fonts with serif, and that monospaced fonts (where the space occupied by a single letter is constant regardless of the actual width of the letter) are more readable than proportional fonts (where space depends on the letter's width). According to the results obtained in reading performance and to subjective preferences, the fonts recommended for children with dyslexia are Helvetica, Courier, Arial, Verdana and Computer Modern Unicode. Arial Italics, instead, seems to be difficult to read on screen. More generally, sans serif and roman font types increased the reading performance, while italic fonts did the opposite [25]. Results from previous studies indicate that subjective readability increases with increasing font size, but that there is a plateau around 18 pt–22 pt, and a decrease beyond 22 pt [26]. Also, subjective comprehension seems to be significantly better for larger font sizes (18, 22, 26 pt) than for smaller ones (10, 12 pt). The same authors, however, also found that smaller line spacing leads to better subjective understanding of the text compared to very large spacing, while intermediate spacing differences seem to have no impact [27]. It should be noted that this study used eye-tracking measures (fixation duration) for readability and no objective measure of accuracy was recorded, nor was any further measure of reading speed. On the other hand, a large study on Spanish subjects with and without dyslexia [28] and an Italian study [29] clearly show that there is no correspondence between the subjects' objective performances and their subjective evaluations and preferences. So-called dyslexia-friendly fonts (DF) like, e.g., Dyslexie,

Open Dyslexia or EasyReading are designed to help people with DD to recognize letters, distinguish between letters of similar shapes, and limit crowding effects. Characteristics of these fonts are a specific letterform, such as increased thickness near the bottom, angling and changing the height and the contours of similarly shaped letters, and increased spacing. Bachmann [30] found that children with DD read a text with a specific font (EasyReading™) faster and with less errors when compared with a standard font like Times New Roman. EasyReading™ was developed for individuals with DD and integrates particular graphic features (e.g., letterform with dedicated serifs, and longer ascenders and descenders) and enlarged inter-letter and inter-word spacing (spacing between two lowercase letters vary between 16 and 18% of the body size, and spacing between two words corresponds to 39% of the body size). This “increased spacing” effect has partially been replicated by Duranovic et al. [31]. Bosnian-speaking children with DD improved in reading accuracy, but not in speed. Other recent studies failed to find any advantage of a dedicated letterform or increased spacing effects comparing DF fonts, Arial, and Times New Roman [32,33]. Other studies did not find any differences between children with and without DD in reading speed and accuracy when they read texts written with DF letterforms or with standard letterforms [32–34]. Both children with DD and typically developing children showed a decrease in reading speed when text was presented with an unusual spacing condition: it seems to be difficult to segment sentences into words when the sizes of the (increased) inter-letter and the (default) inter-word spacing are similar [34]. A significant interaction between letterform and inter-word spacing emerged in the group of children with DD, whose reading speed was similar for a text with DF or with standard letterform and increased spacing [34].

Overall, the studies show great heterogeneity of results, probably based on substantial differences in relevant variables: the age of the participants (children or adults), their characteristics (with/without DD), the type of text used (single words, single letters, nonwords or passages), the parameters with which the outcome was assessed (comprehension of texts or reading speed and/or accuracy, lexical decision, satisfaction questionnaires, eye movements), the type of design and analysis used (manipulation of single or multiple variables, DD/control comparisons vs. differences within groups), the language of the text (orthographically regular languages such as Italian or Spanish or irregular languages such as English). In most studies, group averages are taken into account rather than individual differences. A systematic study conducted on a large number of parameters [29] suggests that the individual effects are limited at the group level and can instead vary from subject to subject (turning out to be facilitating for some and penalizing for others), leading to the conclusion that a highly personalized approach is desirable.

Much less information is available concerning the advantages of various types of TTS (text-to-speech) technologies to support reading and comprehension in children with DD. Better learning [35] and reading comprehension [36] performances have been observed with TTS compared to no TTS for students with reading and language difficulties. However, a recent study suggested that TTS is not necessarily advantageous to all children with reading difficulties, but only for those with typical profiles where listening comprehension is less impaired than decoding skills [37].

Thanks to Italian Law 170/2010, which contains norms and guidelines on educational rights for students with specific learning disorders (SLD), a DD diagnosis brings about the right to use certain compensatory or dispensatory tools at school [38]. Research on SLD has made it possible to create increasingly versatile technologies to support reading, writing and study, and Seleggo is one of these new technologies.

1.3. The Seleggo Platform and the Aim of the Study

Seleggo [39] is a free compensatory tool for dyslexic students. It is an online platform where school textbooks are transcribed and visualized on the screen in PDF versions, characterized by special attention to graphical features and readability, and can be listened to via TTS technology. Special functions such as karaoke modality, conceptual maps

building software, summary creation, verbal and visual definition search, and the possibility to record and retrieve one's own notes and sketches are also provided. Each student with SLD has a personal library and can choose the parameters for text visualization to facilitate reading, modifying the display of the text, choosing font, size, and spacing, as well as TTS parameters, such as speed and pitch.

The availability of many options for text visualization and TTS, however, can be advantageous only as far as the students are able to choose and use parameters that objectively facilitate their reading and comprehension. Since the analysis of the literature, as shown above, suggests that subjective choices may be misleading and that general recommendations may not fit the individual's needs, we decided to build an automatized, as objective as possible, procedure to help the students find the combination of parameters that is most likely to facilitate them in reading and studying. This procedure, composed by an initial, gross-grained subjective procedure and followed by a thorough, systematic analysis of reading and comprehension performance, was named the "Seleggo test" to match the compensatory tool it was meant to complement.

The present study aims to assess the actual advantage of personalized parameter selection through the "Seleggo Test" procedure in supporting reading compared to the use of standard parameters. The working hypothesis is that children with reading disorders would benefit from personalized text features and TTS.

2. Materials and Methods

2.1. Participants

A total of 78 Italian students aged between 8 and 14 years were involved in the study. The sample consisted of 49 children with atypical reading development (AD), either diagnosed with specific reading disorders (SRD) or referred by the schools as special educational needs (SEN), and 29 children with typical reading development (TD) children. All children with SEN were to have specific difficulties in reading, even if not all criteria (severity, specificity, etc.) were met for an SRD diagnosis. Since analysis of performance and profiles in the two groups revealed great overlap, and criteria for distinguishing children with SRD from children with SEN were found to be linked to family compliance and to timely diagnosis more than to objective differences in reading ability (several children initially classified as SEN turned out to have all criteria for being classified as SRD), it was decided to keep the two groups together as a single group defined as AD. The recruitment took place between October 2022 and July 2023. AD children were recruited in either the participating schools or among patients of the neuropsychiatry unit of IRCCS "Eugenio Medea" in Bosisio Parini, Northern Italy, while TD children were recruited in primary and middle schools of the same geographic area. The children and their families were contacted by the researchers, the purpose of the study was explained, and written parental informed consent was collected.

Participants had to fulfill the following inclusion criteria: (a) age between 8 and 14; (b) monolingual speakers or bilingual speakers with sufficient mastery of the Italian language (at least two consecutive years regular attendance of an Italian school). Additional criteria for the AD group were: (c.1) having been previously diagnosed with SRD on the basis of standard inclusion/exclusion criteria [1] or (c.2) having been declared to be a student with special educational needs by the school board. For the TD group, an additional criterion was (c.3) normal school achievement as reported by teachers and parents. The study was approved by the Local Ethics Committee in accordance with the Declaration of Helsinki.

2.2. Measures and Procedure

2.2.1. Procedure

Children were tested individually at their school or at IRCCS "Eugenio Medea". As a first step, each child completed the online Seleggo Test in order to identify video (font, size, spacing) and audio (voice, speed, pitch) personalized parameters. This procedure

is subdivided in a subjective and an objective phase. In the subjective procedure, the child is first of all asked to choose the preferred voice among those supported by their computer. Subsequently, they have to choose one of two font families by clicking on one of two paragraphs representing the same text, each written with a combination of fonts belonging to the same font family (see Figure 1). The three families were comprised of the following fonts, all chosen among Google fonts for maximum compatibility with a variety of text processing systems found on the web: (a) “sans serif” family (automatically selected): Open Sans, Roboto, Cantarell, Overpass, Merriweather Sans, Montserrat, SourceSans Pro); (b) “handwriting/script” family: Amaranth, Montserrat Alternates, Biryani, Muli, Raleway, EasyReading Pro; (c) “with serif” family: Lora, Anonymous Pro, Josefin Slab, Times New Roman, Roboto Mono. In the next step, the three preferred fonts out of each of two families (the family selected by the child and the family described as more dyslexia-friendly in the literature, i.e., some of the most representative sans-serif fonts and some so-called “dyslexia-friendly” fonts, constituting a default-family not participating in the first step of the subjective selection) were chosen by the child. Finally, the computerized (objective) procedure identified, among the 6 fonts selected through the subjective procedure, the specific parameters that allowed each child to read as quickly and accurately as possible, or to understand a text heard. After the objective procedure, such personalized visual and auditory parameters were used in online and offline reading and dictation of nonwords tasks, comparing performances obtained using individualized and standard parameters (standard fonts and the speech synthesis normally used in the Seleggo reader).

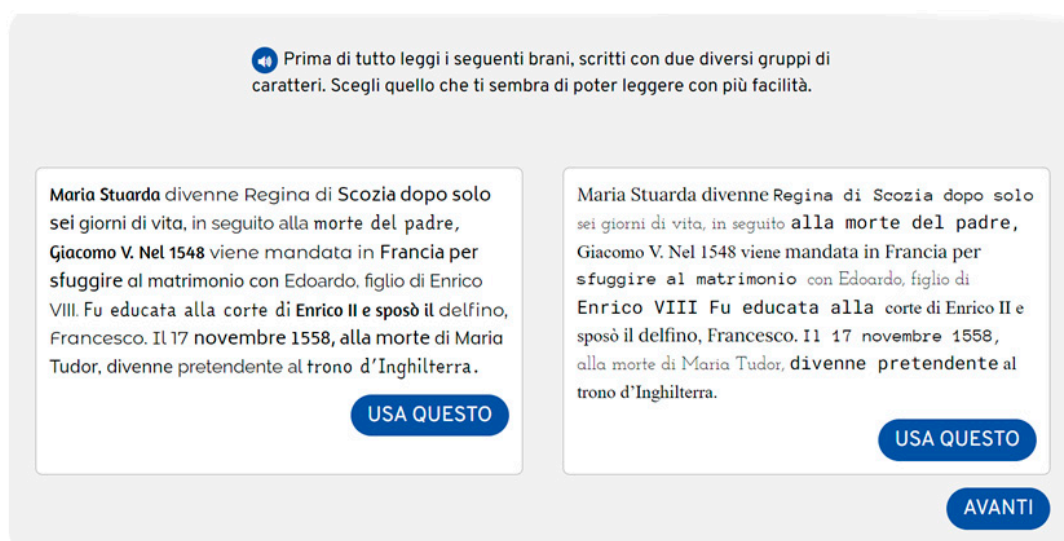


Figure 1. Screenshot of the presentation of the two font families (handwriting/script in the left box and serif in the right box) in the subjective procedure of the Seleggo test. The translations of the instructions in the upper part of the screen and of the text in the blue buttons as shown in the screenshot are the following: “First of all read the following passages, written with two different groups of print characters. Choose the one you feel you can read more easily”. “USA QUESTO” = “Use this one”. “AVANTI” = “forward”. The text in the white boxes is an excerpt of a history book describing Mary Stuart’s life and the textual content of the two boxes is identical except for the fonts used.

2.2.2. Seleggo Test Application

The Seleggo Test was developed to optimize use of the Seleggo platform, an already existing tool providing digitized school textbooks and TTS technology, and allowing each student to choose the parameters for text display, font type, size and spacing, as well as TTS parameters (speed and pitch). The Seleggo Test was designed to offer students the possibility to choose personalized parameters following objective rather than subjective criteria. The Seleggo test procedure is divided into three phases.

- (a) First of all, each child reads different lists of 96 nonwords (perfectly balanced for presence of graphemes, consonants, vowels and their position within the nonword) written in different fonts and sizes (4 sizes for each of 6 different fonts resulting from subjective selection, each combination applied to 4 different nonwords). Accuracy is calculated as the percentage of reading errors (it is possible to have correct, incorrect or self-corrected reading, counting as half, i.e., 0,5 points), while speed is measured in syllables/second. Based on reading speed and errors, the software identifies optimal fonts and sizes.
- (b) Subsequently, these parameters are used to display 4 pseudosentences with different combinations of inter-letter and inter-line spacing. Each pseudosentence is made up of nonwords occupying three to four lines on the screen (depending on spacing parameters). Each pseudosentence is composed of 21 nonwords (2 monosyllabic, 5 bisyllabic, 7 trisyllabic, 5 quadrisyllabic, and 2 pentasyllabic nonwords) summing up to a total of 63 syllables, 39 characters and 5 punctuation symbols (3 commas and 2 full stops). All vowels are represented the same amount of times in all pseudosentences, and so are consonants, but never in the same combination. Accuracy is expressed as the percentage of words read incorrectly, while speed is reported in syll/sec. Based on reading speed and accuracy, Seleggo Test selects the best combination of font, size and spacing for each child. The selected combination is shown on the screen and the child is invited to print it or save it (see Figure 2).
- (c) As a last step, a series of 9 pseudosentences (made up of different non-words) appear on the PC screens and the child is asked to listen to the speech synthesis reading the pseudosentences, with different combinations of speed and pitch. For each pseudosentence, the last nonword read is not displayed on the screen, and the child is asked to choose, among four options, the one corresponding to the nonword heard. The pseudosentences are constructed so as to follow the phonotactic rules of Italian sentences and to suggest the presence of a pseudo-morphosyntactic structure that is completed by the last nonword. Each pseudosentence is composed of three nonwords, one of which is an existing Italian preposition. The second nonword is a bisyllabic nonword whose grammatical category is completely non-transparent. All final nonwords are quadrisyllabic nonwords formed by 3 CV and 1 CVC or CCV syllable, respecting the phonotactic rules of Italian, for a total amount of 9 letters. The different vowels and consonants are equally represented in the nonwords, and their combinations and positions are balanced across conditions. Each of the quadruplets for multiple choice are composed of the target nonwords, one nonword with 3 correct and 1 incorrect syllable, one with 2 correct and 2 incorrect syllables, one with 1 correct and 3 incorrect syllables. The change producing an incorrect syllable affects a single letter of the syllable, which is usually substituted by a phonologically similar element. According to the accuracy of the answer (calculated as the number of correct syllables), the best combination of speed and pitch is selected for each child. Also, in this case, the selected combination is visualized on the screen and the child is invited to either print it or save it for future use.

For both reading subtests (a and b), reading time was automatically recorded by the system through a self-pacing procedure (the child pressed the bar to visualize the following nonword or pseudosentence). Accuracy, by contrast, was judged and recorded by the examiner at the end of each subtest. The system recorded the child's reading and the examiner was required to select for each stimulus whether it was correct, incorrect or self-corrected (after listening to each audio-recording) by selecting the corresponding score on the test correction page (see Figure 3). In case a nonword or a sentence could not be judged, or in case the reading or the recording had been interrupted for any unexpected reason, the examiner could flag the word and request its repetition before continuing with the next steps of the procedure. After all the stimuli had been correctly recorded and scored, the system calculated reading speed expressed as syllables by second, for each

of the parameter combinations (font and size in a, spacing in b). Accuracy of part c is calculated automatically by the software.

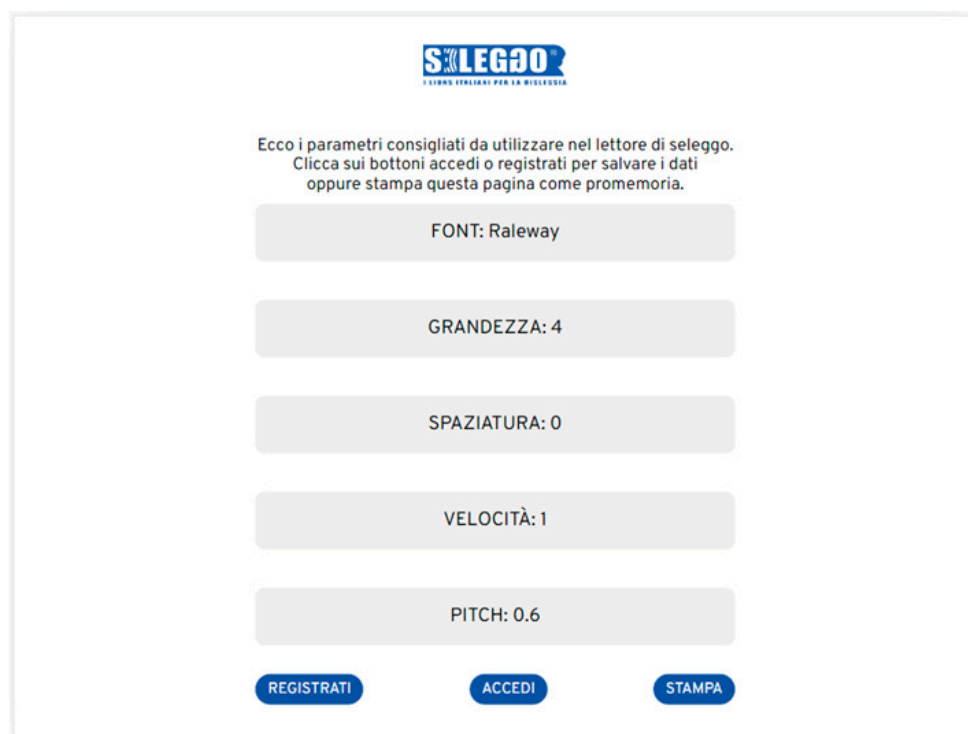


Figure 2. Screenshot of the recommendations for font, size, spacing, speed and pitch as presented on the screen at the end of the first and second phase of the Seleggo test. The translations of the instructions (upper part of the screen) and of the information reported in the screenshot are the following: “Here are the parameters suggested to be used in the Seleggo reader. Click on the buttons “access” or “register” to save your data or print this page as a reminder”. “GRANDEZZA” = “size”; “SPAZIATURA” = “spacing”; “VELOCITÀ” = “speed”. The text in the blue buttons at the bottom of the screen can be translated as follows: “REGISTRATI” = “register”; “ACCEDI” = “access”; “STAMPA” = “print”.

2.2.3. Seleggo Test Validation

The selected personalized visual and auditory parameters were compared with the standard ones in order to verify the validity of the procedure. A first part of this procedure was completely automatized and was delivered by the same platform offering the Seleggo Test. Each child had to read a text on the PC screen, divided into 24 sentences (8 visualized in the font that had been selected through the Seleggo Test, 8 sentences visualized in standard font-1, i.e., Times New Roman and 8 in standard font-2, i.e., Roboto Mono). Size and spacing for the personalized font were the ones selected through the Seleggo test, while for the two standard fonts they were the default parameters found on the Seleggo reader, corresponding to size “1” and spacing “1”). The fonts were presented in an alternating, random sequence. Speed was recorded as the number of syllables read in a second for each of the chosen fonts, while accuracy, recorded by the examiner as in the previous subtests, was expressed as percentage of errors for each font.

After that, the procedure continued on the Seleggo reader: the child was asked to listen to 4 lists of 8 bisyllabic nonwords each read by TTS (2 lists in the chosen voice and 2 lists in the standard voice = Google Elsa) and write them one by one on a sheet of paper. The lists of nonwords in the two groups (to be read by the standard and the personalized voice, respectively) contained exactly the same syllables but the exact combination of syllables in the nonwords differed and the position of each syllable in the nonword was counterbalanced across groups. All the nonwords followed the phonotactic rules of the

Italian language but they were not similar to any existing word (nonwords that had orthographic neighbors were excluded). Accuracy was measured as the number of errors made in writing the nonwords dictated by each voice (maximum one error was counted for each nonword).



Clicca sul tasto play a sinistra della traccia per riascoltare la lettura dello studente e poi indica se la parola era corretta (al primo tentativo), errata oppure autocorretta (prima letta in modo sbagliato e poi in modo corretto, senza che sia stato segnalato l'errore).

	Accuratezza	Tempo di lettura	Da rileggere
► Mefrito	corretto	1.56 sec	☐
► Brizefo	errato	0.479 sec	☐
► Stidoma	auto corretto	1.019 sec	☐

Figure 3. Screenshot of the examiner interface for assessment of reading accuracy after the nonword reading test (reading time is recorded automatically by the system). The instructions in the upper part of the screenshot read: “Click on the “play” button to the left of each trace in order to listen to the student’s reading. Then, indicate whether the word has been read correctly (at first try), incorrectly or self-corrected (wrong in the first try and corrected immediately afterwards, without any alert about the error).” The titles of the table shown in the screenshot are translated as follows: “Accuratezza” = “accuracy”; “Tempo di lettura” = “reading time”; “Da rileggere” = “to be read again”. The first column on the left shows the nonwords that were shown to the child; by clicking on the “play” button, the examiner can listen to the recording of the child’s reading.

2.3. Data Analysis

First of all, in order to assess the validity of the measures collected through the Seleggo test, the performances of atypically developing children (AD) and typically developing (TD) children were compared for each of the collected variables, including speed and accuracy of the nonword reading task, of the pseudosentence reading task, and the selection of nonwords uttered by the TTS program. Due to the skewness of some of the reading variables (especially related to accuracy), nonparametric tests (Mann–Whitney) were preferred. Since the attended grades of the children belonging to the two groups were practically identical (t -test, $p = 0.95$), nonparametric tests were applied on raw scores without taking into account the influence of grade. Group-related differences in the choice of parameters were assessed with chi-square, Lambda (indicating the degree of predictability of the choice based on group) and Goodman–Kruskal’s tau (indicating the degree to which information about group can improve prediction of the choice of parameters) in the case of purely categorical variables such as font type or voice. When the dependent variable could be considered as an ordinal variable (for instance font size, spacing or voice speed and pitch) Somers’ D (a non-parametric index used to evaluate the asymmetric association between two ordinal qualitative variables) was preferred.

As a second step, the results of the validation procedure were analyzed, comparing the performances obtained by the children (both as a whole group and specifically for each group with and without reading difficulties) with the standard fonts and voices to those obtained applying the selected visual and auditory parameters. The Wilcoxon Z test was applied to assess differences between the standard and personalized parameters within each group. Additionally, a variable expressing the advantage of the personalized script/voice was calculated subtracting the measures of speed (syllables per second) and

accuracy (number of errors) of the standard (average of the two standard fonts) from the personalized condition. This means that a positive measure expresses the advantage of the personalized over the standard condition for speed (more syllables in one second) and a negative measure expresses the advantage for accuracy (fewer errors).

Finally, data from the offline procedure were analyzed, again comparing the results for standard and personalized conditions. One-tailed p -values were considered due to a priori, unidirectional hypotheses: an advantage of the personalized condition was expected. Also in this case, speed (in total time) and errors obtained with (averaged) standard fonts were subtracted from those obtained with personalized fonts, respectively, and the same was conducted for accuracy measures in the dictation task, subtracting errors made when the nonwords were dictated by the standard voice from errors made when the personalized voice was dictating. Again, a negative value in these variables indicates an advantage of the personalized over the standard condition: less time, fewer errors.

As a last step, differences between personalized and standard conditions were compared between TD and AD groups. The distributions of difference-variables were plotted and the variables related to reading (especially accuracy) revealed large variance spanning over both positive and negative values. The distribution of the original variables was more linear but some of the variables showed deviation from normality. For these reasons, non-parametric tests were preferred for the analyses. Two-tailed tests were considered in this case, since no clear a priori hypotheses had been formulated concerning group differences.

3. Results

All the main variables expressing reading speed and accuracy (errors) as well as the ability to select the nonword pronounced by the TTS software showed significant differences between typical and atypical readers. These differences, shown in Table 1, confirm the validity of the computerized task, in terms of its ability to capture the differences between students with and without known reading difficulties.

Table 1. Descriptive statistic of TD and AD groups and results of groups comparison. Significant differences are shown in bold.

	TD Children <i>n</i> = 29		AD Children <i>n</i> = 49		Group Comparison
	Median	Range	Median	Range	Mann–Whitney U, <i>p</i>
TTS matching accuracy (mean syllables)	3.93	3.59–4	3.78	2.96–4	416, 0.004
Word reading speed (syll/s)	1.34	0.73–1.67	1.07	0.47–1.56	390, 0.001
Word reading accuracy (errors)	7.30	0.50–45.8	19.27	2.1–53.1	282.5, <0.001
Pseudosentence reading speed (syll/s)	1.62	1–2	1.30	0–2	405.5, 0.002
Pseudosentence reading accuracy (errors)	4.82	0.51–31.22	11.93	1.52–59.14	285, <0.001

3.1. Results of Seleggo Test

The results of the personalization procedure are described in the following sections.

3.1.1. Visual Parameters

Figure 4 shows the distribution of the font types selected by the two groups (TD, AD) as a result of the personalization procedure. It can be observed that the variety of selected fonts is larger for the TD (14 different fonts) than for the AD group (11 different fonts), even if the AD group comprised more participants. The most frequently selected fonts are

Biryani and Merriweather Sans for both groups, followed by Montserrat and then Overpass for the TD group, whereas the AD group lists Overpass and then Raleway. Easy Reading and Muli are equally frequently selected both in the TD (6.80%) and in the AD (8.16%) group. Specific statistics to assess asymmetry in measures of association or the possibility to predict group from font selection did not yield any significant result (chi-square, Lambda and Goodman–Kruskal's tau all $p > 0.05$).

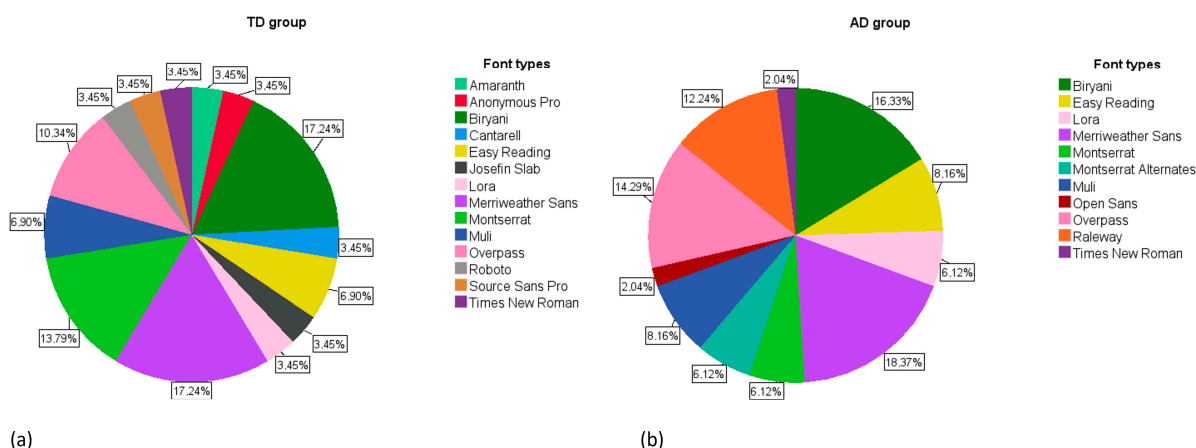


Figure 4. Descriptive results of font type selection outcomes are shown for the TD group (a) and for the AD group (b) separately.

Figure 5 shows the distribution of the parameters size and spacing in the two groups. Results show that 2 was the most frequently selected size for both groups, but a higher percentage of children in the AD compared to the TD group turned out to read better with larger sizes. As to the spacing parameters, the AD group obtained better results with space 4 (40.82%), while the TD group was more facilitated by space 1 (34.48%). The statistics predicting size and spacing from group (Somers' D) yielded a significant result for size (Somers' D = 0.273, $p = 0.013$) but not for spacing ($p = 0.436$).

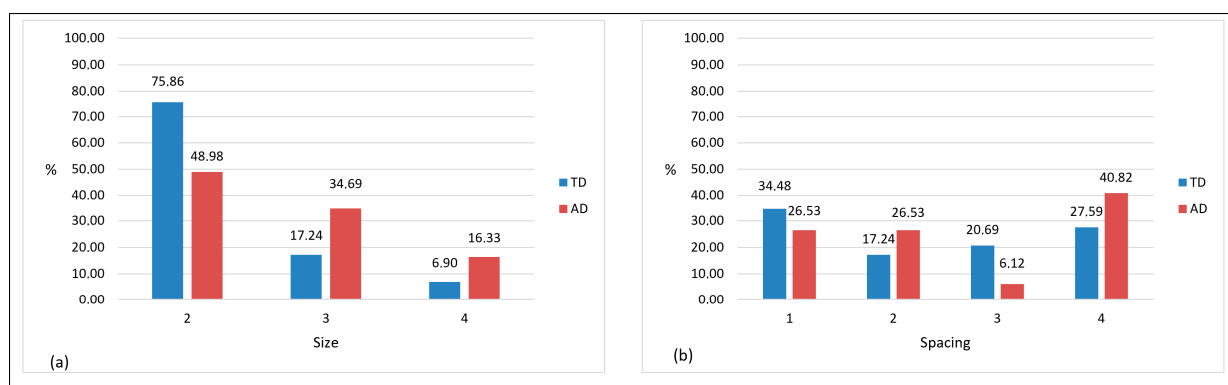


Figure 5. Outcomes of the objective procedure for the personalization of font size (a) and spacing (b), for the two groups (TD and AD).

3.1.2. Auditory Parameters

Figure 6 shows the auditory parameters (speed and pitch for the TTS voices) resulting from the objective personalization procedure in the two groups. The two groups did not show fully significant differences concerning speed (Somers' D = 0.150, $p = 0.054$), with both groups apparently facilitated by slower speed (0.8) and by a lower pitch (0.6) but only the differences concerning pitch selection were significant (Somers' D = 0.143, $p = 0.005$), showing only some children from the AD group obtaining better results with pitch 0.8 and 1.0) (see Figure 6).

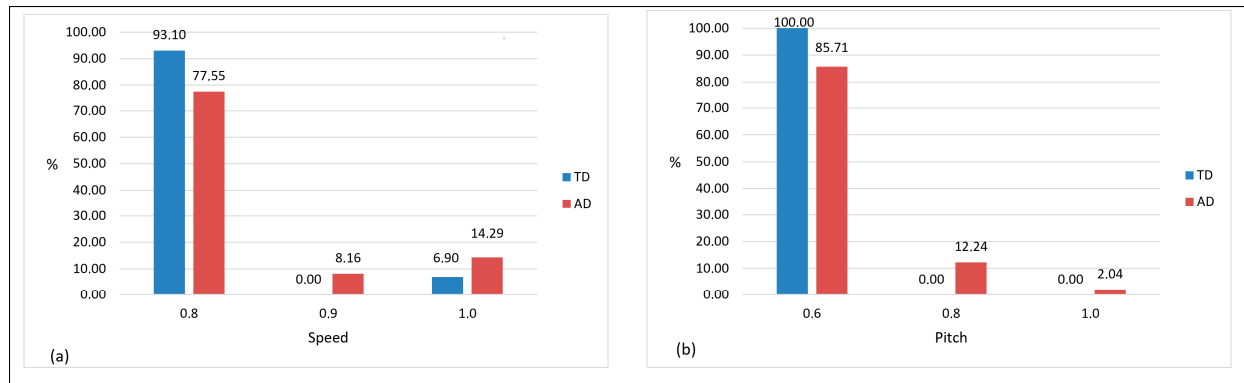


Figure 6. Outcomes of the objective procedure for the selection of TTS voice speed (a) and pitch (b), for the two groups (TD and AD).

3.2. Results of the Validation Tests

As to the results of the validation of the personalization procedure, the data concerning the reading tests were analyzed comparing the scores obtained with the personalized visual parameters and those obtained with the standard parameters. First of all, the analyses were performed on the whole sample (Table 2, Figures 7, 8 and 9a).

Table 2. Results of validation procedures and comparison between personalized and standard visual parameters in the TD and AD groups. Significant results are shown in bold.

Whole Sample ($n = 78$)			
		Median (Range)	Wilcoxon Test Z (p , 1-Tailed)
Text reading speed (syll/s)	Seleggo	2.85 (0.51–4.78)	−1.66 (0.049)
	Standard	2.85 (0.54–4.70)	
Text reading accuracy (errors %)	Seleggo	1.63 (0–29.32)	−1.91 (0.028)
	Standard	2.69 (0.48–22.92)	
Dictation accuracy (errors)	Seleggo	9 (1–13)	−1.65 (0.0495)
	Standard	9 (4–15)	

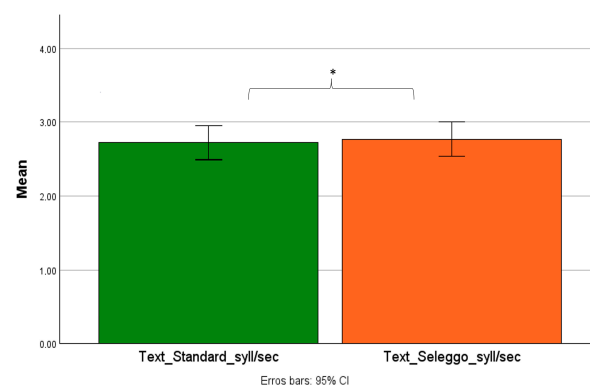


Figure 7. Text reading speed (syll/s): comparison of Standard and Seleggo parameters (* indicates significant differences at $p < 0.05$, 1-tailed).

Subsequently, the facilitation effects (i.e., the differences between performances with personalized and with standard parameters) were compared in the two groups with and without reading difficulties. None of the differences turned out to be statistically significant (all $p > 0.733$, with exception of differences related to writing errors, which, however, did not reach significance, $p = 0.210$). Therefore, comparisons within groups were not

further computed, but the advantage of personalized over standard voice in writing to dictation was separately analyzed in the two groups (Figure 9b). Indeed, while there was no significant advantage of personalization in the TD group (Wilcoxon $Z = -0.09$, $p = 0.463$, 1-tailed), this effect was clearly significant in the AD group (Wilcoxon $Z = 2.05$, $p = 0.020$, 1-tailed).

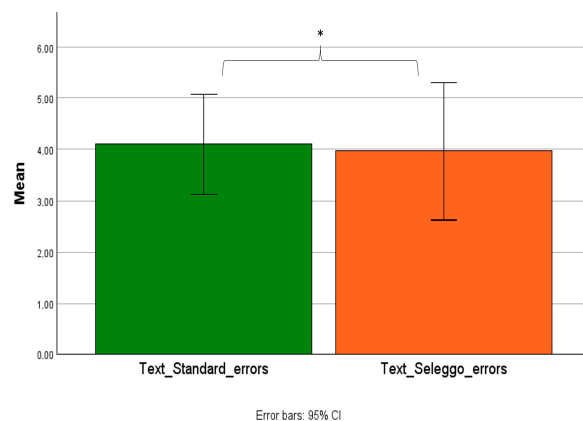


Figure 8. Text reading errors: comparison of Standard and Seleggo parameters (* indicates significant differences at $p < 0.05$, 1-tailed).

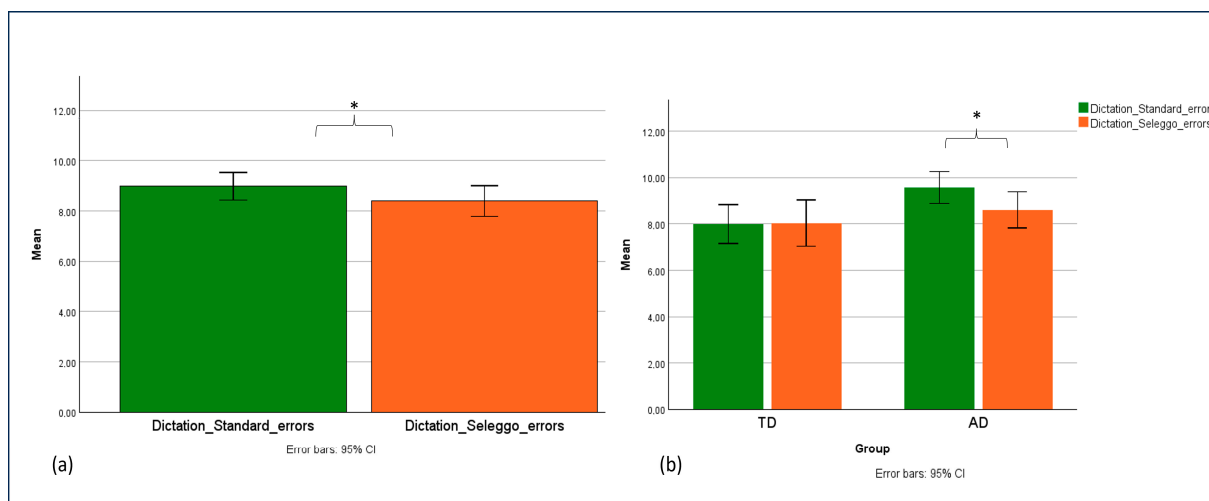


Figure 9. Dictation errors in the whole group of children (a) and in TD e AD groups (b) separately (* indicates significant differences at $p < 0.05$, 1-tailed).

4. Discussion

The aim of the present study was to assess the validity of a special automatized procedure, called the “Seleggo Test”, designed to select the most favorable parameters for both text visualization and text-to-speech conversion. The selected parameters can be used for the Seleggo application [39], supporting reading and study, as well as with any text processing software allowing for personalization. The hypothesis was that children with reading disorders could benefit from a personalized font and readers.

The designed procedure for personalization follows both a subjective gross-grained approach and, due to clear prior evidence that subjective choices do not reflect objective advantage in reading [25,28,29], an objective, systematic comparison of the application of different parameters in reading and listening efficiency.

In the first part of the study, in order to preliminarily assess the discrimination capacity of the measures collected through the Seleggo test, performance of AD and TD children were compared. Results showed that the scores obtained through the Seleggo test procedure

differ significantly between typical and atypical readers. In particular, reading performance of TD children was faster and more accurate, as was the selection of non-words pronounced by the TTS software.

Subsequently, the results of the validation procedure were analyzed, comparing the performances obtained by the children with the standard fonts and voices to those obtained applying the selected visual and auditory parameters, within each participant. Results of the whole sample showed significant facilitations of personalized parameters in reading speed and accuracy, and in accuracy in writing to dictation.

The facilitation effect was similar in the two groups for all parameters. A post hoc check of the effects produced in the two separate groups for writing to dictation revealed that AD children were more accurate in the dictation task adopting the Seleggo test parameters, whereas TD children did not show such large facilitation effects.

As found in previous studies (e.g., [21–23,27]), font types were confirmed to have a significant impact on readability for both children with and without reading disorders. It is immediately evident that there is no single font type giving a clear advantage in reading for children with AD. The range of selected font types is wide for both groups (although it is admittedly less wide in the AD group, where a total of 11 different font types were selected as optimal font versus 14 in the TD group). The most frequently selected fonts do not reach 20% of total outcomes in both groups. In the present study, children of the TD and AD groups chose above all Biryani (17.24% and 16.33% in the TD and AD group, respectively) and Merriweather Sans fonts (17.24% and 18.37%), both sans serif fonts (although Biryani was classified as a handwriting/script type due to some changes in letters such as, e.g., the lowercase l (“el”), becoming taller and different in form compared to the uppercase I (“i”). Taking into consideration font size and spacing, TD and AD children behave rather similarly, even if 75% of the TD group chose size 2, while children of the AD group are more distributed in the three dimensions of the font. Moreover, most of the TD group chose spacing 1, while in the AD group the spacing allowing for the best reading performances was 4. The descriptive data of the present study point to the fact that children without reading disorders also prefer, and read better, the fonts that people with dyslexia prefer [25]. As to TTS parameters, both TD and AD children performed better with slowed speed (0.8) and lower pitch (0.6), although the non-generalizability of these principles to all individuals was evident, especially in the AD group.

So-called DF fonts, such as EasyReading™, were not the most frequently selected font types, but they were selected by 5 to 10% of the children in both groups. Very common font types such as Times New Roman and Roboto were selected very rarely (2–3% in both groups), a result that came rather unexpected. This actually confirms the results by Bachmann and Mengheri [40] showing significant differences in reading fluency and accuracy for EasyReading™ compared to Times New Roman.

More generally speaking, the aim of the present study was not to compare specific fonts nor to establish any ranking. As expected, the study shows that each child improved in reading speed and accuracy and in dictation accuracy when using a font customized to their needs. The Seleggo test confirmed the expectations to be a reliable and useful tool allowing for the identification of the best parameters for each child.

Regarding voice parameters, children in both the TD and the AD group performed better with speed 0.8 (slightly relented with respect to natural voice) and a 0.6 pitch (a deeper, less acute pitch compared to the default voice). Research that examined the use of TTS as a compensatory tool found better reading comprehension performances with TTS compared to no TTS for students with reading and language difficulties [35,36]. This supports a general recommendation that the use of Assistive Technology to support reading and study in DD can be of great help. Even if very little data are reported in the literature concerning objective advantages in the use of TTS to enhance reading accuracy, the present results suggest that personalized TTS parameters may further facilitate text comprehension.

5. Limitations of the Study and Future Directions

There are some limitations to the present study. First of all, the subjective procedure introducing the selection of fonts and sizes biases the absolute probability for each single font type to be selected (since the sans-serif family has been often indicated as favoring reading in dyslexia, it was automatically selected as a default option; by contrast, only one of the two other font families had to be selected directly by the child). This may have led to an over-representation of fonts belonging to the sans-serif family in the final distribution, so that any direct comparison of different fonts (which indeed was beyond the scope of the study) could be impacted by this initial bias. Nonetheless, comparisons within each font family as well as comparisons between groups of children are not affected and can be considered completely informative. Second, it should be highlighted that the results of the study have been analyzed comparing children with typical reading development and with atypical reading development, irrespective of diagnosis. The Atypical Development group thus included both children with full diagnosis of specific reading disorder and children with special educational needs, whose performance profiles were found to be rather similar. This implies that the outcomes of the study should not be considered as strictly specific to children with Dyslexia but extend to children with reading disorders who do not have a diagnosis. Therefore, further research could focus on specific advantages in the use of personalized parameters by applying more stringent criteria to distinguish children with SRD from children with SEN. A further interesting application of the test could concern children acquiring a new language, especially when new characters are introduced (for instance, children who have learned to read Cyrillic, Arabic or logographic scripts and need to familiarize with Western script).

The Seleggo Test has been specifically developed as an objective procedure to optimize the choice of font type, size and spacing, as well as TTS parameters (speed and pitch) for use of the Seleggo platform, where support for reading and studying is provided also through other compensatory tools. However, the test as well as the platform have been developed only for the Italian language, and this could constitute a further limitation of the study. Nonetheless, an adaptation of the platform (including the test) to other languages is envisaged and could be realized with limited effort, so as to extend the possible applications of the present results. Moreover, the indications obtained through the test can be used to facilitate everyday reading activities, personalizing access to written text both on screen and, if necessary, in print or in voice, through any text processing or TTS application. Last but not least, the information obtained on the parameters facilitating and hampering reading for each individual child could be used for the programming of rehabilitation procedures that focus on the visual and auditory characteristics of the input to be coded/decoded (see, e.g., [20,22,41]), thus enhancing the personalization of intervention strategies.

6. Conclusions

Overall, the results of the study confirm the advantage of using personalized parameters for online reading and for TTS systems, and the usefulness of objective procedures to integrate subjective choices to select the best parameters based on actual facilitation rather than on aesthetic and pleasantness reasons. The advantages of personalization can be observed in reading speed but are especially evident in the reduction of reading and writing errors. Altogether, the results of the present study add to mounting evidence about great heterogeneity within the group of children with reading disorders, and the consequent need for personalization of learning strategies, tools and pathways to support literacy acquisition and, more generally, access to knowledge in this population.

Author Contributions: Conceptualization, M.L.L., M.M., M.C. and P.P.; methodology, M.L.L.; software, G.P. and A.C.; validation, M.L.L., P.P. and F.B.; formal analysis, M.L.L. and F.B.; investigation, P.P., G.B. and M.C.; resources, M.M.; data curation, M.L.L., G.B. and F.B.; writing—original draft preparation, F.B. and M.L.L.; writing—review and editing, M.L.L. and F.B.; visualization, F.B.; supervision, M.M.; project administration, M.M.; funding acquisition, M.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Italian Ministero della Salute, grant number RC2023.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki, and approved by the Institutional Review Board (or Ethics Committee) of Scientific Institute IRCCS E. Medea (protocol code Prot. N. 62/22—CE, approved on 22 October 2022).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are openly available in Zenodo, <https://doi.org/10.5281/zenodo.10205803> (accessed on 7 January 2024).

Acknowledgments: The Authors wish to thank Seleggo NPO for supporting the research project by funding M.C.'s work for the validation of the Seleggo test app. They also wish to thank the schools who participated in the study: Giovanni Testori Comprehensive Institute in Novate Milanese (schools Maria Montessori, Italo Calvino and Orio Vergani).

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results. The author (M.C.) who received direct funding from Seleggo NPO is not part of the Seleggo organization; her involvement mainly concerned the design of the testing procedure and her contribution to data collection was limited to less than 10% of the total sample. Betterdays Ltd. was involved in the technical development of the Seleggo app but it does not own the application and has no commercial interests in its use and diffusion, nor does it have any other financial or non-financial conflicts of interest. The Seleggo web app is owned by Seleggo NPO (part of the Lions Club International association) and is available free of charge for registered schools and for individual students with SRD or SEN.

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